

Research Statement

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I am a macroeconomist. My research asks how financial conditions, and monetary policy in particular, change the investment decisions of individual firms, and what those decisions mean for long-run growth. I work with large firm-level panels and dynamic econometric methods, and I build structural models to interpret what the data show. My current work focuses on innovation, the form of investment most directly tied to technological progress and also the one most exposed to financing frictions.

Job market paper: Monetary Policy and Firm Innovation

Monetary policy is usually treated as a short-run stabilization tool. A growing literature argues that it can also have persistent effects through investment and innovation. Most of that evidence is aggregate, and it leaves three questions open. Does monetary policy draw new firms into innovation, or does it change how much existing innovators invest? How does a change in R&D spending show up in the quantity and quality of what firms invent? And is the response symmetric across easing and tightening?

To answer them, I link Compustat financial data, USPTO patent records and CRSP market data at the firm level and estimate the dynamic effects of monetary policy shocks with local projections, combined with Probit and Tobit models for the two margins of R&D investment.

The paper has four main findings. First, the R&D response operates almost entirely on the intensive margin. Expansionary shocks raise R&D and knowledge capital among firms that already innovate, and the response is an order of magnitude larger than on the extensive margin, which cannot be distinguished from zero at any horizon. Second, the composition of innovation output changes. After an expansionary shock, patent counts fall in the short run while citations, patent importance and the share of patents in the upper tail of the importance distribution rise over the medium run. Third, a two-stage analysis shows that this reallocation runs through the R&D response itself rather than through a direct effect of the shock on patenting. Fourth, the effects are asymmetric in sign. Contractionary shocks reduce both the quantity and the quality of innovation, and the reallocation toward quality is identified most cleanly among economically large shocks.

These results contribute to several literatures. For work on monetary policy and heterogeneous firms, they show that for innovation the decisive dimension of heterogeneity is whether a firm is already doing R&D, which matters more than the leverage, liquidity, age or size distinctions emphasized for physical investment. For work on monetary policy and growth, they identify a firm-level mechanism through which policy can change the direction of technological progress as well as its pace. For the innovation literature, they show that measures of patent quality do not move together: citation-based and text-based measures rise after an expansionary shock, while per-patent market value shows no reliable response. The conclusions one draws about innovation quality therefore depend on which proxy one uses.

Work in progress

Fixed Costs in Lumpy Investment Models. My job market paper finds that the innovation response to monetary policy runs through the intensive margin. For physical capital, existing work finds the opposite, with the extensive margin central to the investment channel. This project studies the physical side directly. I build a dynamic model of heterogeneous firms that face capital adjustment frictions and observe only a noisy

signal of their own productivity. I solve it by value function iteration, calibrate it to observed investment patterns, and use Monte Carlo simulations to trace transition dynamics after macroeconomic shocks. The goal is to understand how fixed adjustment costs and imperfect information together shape which firms invest after a shock.

Skill demand in AI job postings. This project uses natural language processing and supervised machine learning on millions of AI-related job postings to classify roles, extract required skills and track employer demand by industry and region. It shows where demand for AI skills is spreading in the labor market, the hiring side of the technology questions my other work asks about investment.

Future research

A model of the quality margin. The reallocation from incremental to high-impact research is the central empirical result of my job market paper, and it calls for a model in which firms choose between project types under financial frictions. I plan to build a heterogeneous-firm model in which easier financing lowers the cost of long-horizon, higher-risk projects, and to use it to ask how much of the long-run effect of monetary policy on productivity runs through the direction of research rather than its volume.

Why quality measures disagree. The gap between citation-based and market-value measures of patent quality is a finding in its own right. I plan to study when the two diverge and what each captures, since the answer matters for any work that uses patents to measure the returns to innovation policy.

Technology adoption and financial conditions. Combining the job-posting data with firm financial records would let me ask whether credit conditions affect how fast firms adopt new technologies such as AI, connecting the adoption margin to the investment margins I study in my job market paper.

Across these projects I want to know how the financial environment changes what firms invest in, since the direction of investment turns out to move even when its volume does not.